

Local Dephasing in a de Sitter Static Patch: Sharp Localization–Energy Cost of a Compact Pointer Channel

Brian Fioca

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Abstract

We consider a degenerate pointer qubit coupled through a smooth compact spacetime source to the conformally coupled massless scalar in a four-dimensional de Sitter static patch. The exact Weyl-displacement channel and its thermal dephasing factor are known. We eliminate the source strength and solve the resulting finite-support energy optimization. If the source is confined to areal radius $a < R$ for static time T , set $y = \operatorname{arctanh}(a/R) + T/R$. The halved diamond distance from complete pointer dephasing obeys

$$\log \frac{1}{2\epsilon_{\text{obs}}} \leq E_K R C_{\text{opt}}(y), \quad C_{\text{opt}}(y) = 2y\Lambda(y/2) \leq F(y), \quad F(y) = \frac{4 \operatorname{arsinh}(1)}{\pi} y + \frac{8}{\pi^3} y^2.$$

Here $\Lambda(\tau)$ is the simple top eigenvalue of the positive compact kernel $\pi^{-1} \log[\sinh(\tau(u+v))/\sinh(\tau|u-v|)]$ on $(0, 1)$, and E_K is the exact post-switch scalar Killing-energy excess generated by the same source. The optimization includes arbitrary angular and canonical data; its unique extremizing sector is s-wave momentum. Temperature corrects the small-support vacuum cost only at cubic order, while $8y^2/\pi^3$ is the sharp leading large-support coefficient. Smooth compact sources approach the optimum when their spatial worldtube approaches the final support ball. For flux-free spherical final data, the same scalar datum gives exact vanishing-extrinsic-curvature Einstein–scalar Hamiltonian and momentum constraint data with mass $m = E_K$. We also prove that the dephasing functional is incomparable with the local coherent-state entropy entering recent Klein–Gordon Bekenstein bounds. The result concerns field energy and final-slice constraints. It does not include the material stress of an autonomous switching device, derive the channel on the perturbed geometry, or solve the coupled Einstein–matter evolution.

1 Introduction

A localized quantum reference or observer is not only an abstract register. It must couple to fields for a finite time and within a finite region, and the resulting field state carries energy and stress. This elementary observation is often hidden when an ideal observer is represented directly by a dephasing map. Our purpose is to quantify it in one continuum model where the channel, the field stress, and the causal support can all be treated without a weak coupling expansion.

Harlow, Usatyuk, and Zhao motivate an ideal observer by a quantum-to-classical channel that removes off-diagonal blocks in a stable pointer basis [1]. We use complete pointer dephasing as that operational target. Our channel distance below is not their gravitational encoding error, and the qubit in this model is not identified with their observer entropy S_{ob} .

Gapless two-level detectors linearly coupled to free fields provide exactly such a model. Their time-ordered evolution is an exact Weyl displacement, because free-field commutators are c-numbers. Landulfo used this fact for relativistic communication channels [2]; Barcellos and Landulfo derived the corresponding field-energy and switching-work accounting [3]. More recently, this solvable detector has been used for a

nonperturbative treatment of Danielson–Satishchandran–Wald (DSW) decoherence [4]. Killing-horizon decoherence itself, its local two-point-function formulation, and its de Sitter specialization are by now well developed [5, 6, 7, 8].

Those results determine the dephasing functional for a chosen source. They do not by themselves eliminate the source and ask how large that functional can be at fixed post-switch field energy and finite causal support. This is the question addressed here. In the conformal scalar static-patch model, the answer reduces to a localized comparison between negative powers of the one-particle Hamiltonian and the energy norm. The low-frequency thermal enhancement is dangerous globally, but compact support turns the relevant compressed inverse operators into bounded forms.

Let R be the de Sitter radius, let a smooth source be supported inside a centered areal ball of radius $a < R$ for a static-time interval of length T , and let ϵ_{obs} denote the halved diamond distance between the resulting pointer channel and complete pointer dephasing. Our main result is

$$\log \frac{1}{2\epsilon_{\text{obs}}} \leq E_K R C_{\text{opt}}(y) \leq E_K R F(y), \quad y = \text{arctanh}(a/R) + T/R, \quad (1)$$

where

$$C_{\text{opt}}(y) = 2y\Lambda(y/2), \quad F(y) = \frac{4 \text{arsinh}(1)}{\pi} y + \frac{8}{\pi^3} y^2. \quad (2)$$

Here $\Lambda(\tau)$ is the top eigenvalue of the explicit positive compact kernel

$$k_\tau(u, v) = \frac{1}{\pi} \log \frac{\sinh[\tau(u+v)]}{\sinh[\tau|u-v|]} \quad \text{on } L^2(0, 1). \quad (3)$$

The field energy E_K is generated by the same source that defines the channel. No detector coupling remains in equation (1). The estimate holds for the full conformal scalar, not only its spherical sector. The exact optimizer is the unique positive top eigenfunction of equation (3) in the s-wave momentum sector. A resolvent expansion shows that the angular potential improves the complete thermal kernel, while explicit momentum lower bounds dominate the field-coordinate sector for every $y > 0$.

Largest-eigenvalue and isoperimetric questions for Riesz-potential operators have an established spectral literature, including spherical and hyperbolic geometries [9]. We do not claim novelty for the compact-operator variational principle or its Perron–Frobenius conclusion. The candidate contribution is the exact reflected KMS kernel selected by the detector problem, the reduction of all angular and canonical sectors to that one operator, and the resulting operational support asymptotics.

The exact coefficient has controlled support asymptotics. Globally,

$$\max \left\{ \frac{3y}{\pi}, \frac{8y^2}{\pi^3} \right\} \leq C_{\text{opt}}(y) \leq F(y). \quad (4)$$

The ratio between the analytic upper and lower linear coefficients is $4 \text{arsinh}(1)/3 \approx 1.17516$. More sharply, $C_{\text{opt}}(y) - 2y\Lambda(0) = O(y^3)$ with a rigorous remainder $y^3/(6\pi)$, and $C_{\text{opt}}(y) = 8y^2/\pi^3 + O(1)$. Thus the linear small-support law and the quadratic large-support law are both intrinsic rather than artifacts of the original one-sided estimate.

The relation to gravity is deliberately limited but exact. In the spherical subclass one may finish with vanishing field coordinate and nonzero field momentum. The momentum density then vanishes. The optical Killing-energy measure is exactly the mass measure in the Einstein–scalar Hamiltonian constraint with vanishing extrinsic curvature, giving a necessary radial metric cost and an explicit nonempty local weak-constraint window. The nonzero scalar momentum means the full matter data are not time-reflection symmetric. This is a theorem about initial data on the final post-switch slice, not a derivation of the detector channel on its own perturbed geometry.

There is also a close conceptual comparison with Bekenstein-type bounds. Hollands, Longo, and Morsella recently proved a model-independent local-entropy bound for localized Klein–Gordon wave packets [10]. Their modular entropy and our pointer dephasing exponent are different quadratic forms. We give two radial packet sequences showing that neither uniformly bounds the other at fixed radius. Thus equation (1) should be read as an operational dephasing analogue, not as the first localization–information–energy inequality in QFT.

The paper is organized as follows. Section 2 fixes the static-patch model and recalls the exact channel and stress identities. Section 3 proves the sharp all-angular localization–energy theorem. Section 4 gives explicit trial profiles and smooth source realization. Section 5 constructs the final Einstein–scalar constraint data. Section 6 compares the result with local entropy and horizon decoherence. Section 7 states the claim boundary and open problems.

2 Static-patch model and exact pointer channel

2.1 Optical partial waves

In static coordinates, four-dimensional de Sitter space has

$$ds^2 = -N(r) dt^2 + \frac{dr^2}{N(r)} + r^2 d\Omega_2^2, \quad N(r) = 1 - \frac{r^2}{R^2}, \quad 0 \leq r < R. \quad (5)$$

Introduce the optical radius

$$x = R \operatorname{arctanh}(r/R), \quad r = R \tanh(x/R), \quad \Omega(x) = \operatorname{sech}(x/R). \quad (6)$$

Then $g_{\text{dS}} = \Omega^2 g_{\text{opt}}$, where

$$g_{\text{opt}} = -dt^2 + dx^2 + R^2 \sinh^2(x/R) d\Omega_2^2. \quad (7)$$

For the conformally coupled massless scalar, spherical-harmonic decomposition and the standard radial rescaling give the positive operators

$$A_\ell = -\frac{d^2}{dx^2} + \frac{\ell(\ell+1)}{R^2 \sinh^2(x/R)}, \quad h_\ell = A_\ell^{1/2} \quad (8)$$

on $L^2(\mathbb{R}_+)$, with the regular Dirichlet condition at $x = 0$. The Bunch–Davies state restricted to the static patch is KMS at

$$\beta = 2\pi R \quad (9)$$

for the static Killing flow. We use this standard identification throughout.

Let $(q_{\ell m}, p_{\ell m})$ denote real final Cauchy data for the rescaled partial waves. For smooth compact data, the thermal quadratic form and the Killing energy are

$$\Gamma_{\ell m} = \langle q_{\ell m}, h_\ell \coth(\beta h_\ell/2) q_{\ell m} \rangle + \langle p_{\ell m}, h_\ell^{-1} \coth(\beta h_\ell/2) p_{\ell m} \rangle, \quad (10)$$

$$E_{\ell m} = \frac{1}{2} \left(\|h_\ell q_{\ell m}\|^2 + \|p_{\ell m}\|^2 \right). \quad (11)$$

We write $\Gamma = \sum_{\ell m} \Gamma_{\ell m}$ and $E_K = \sum_{\ell m} E_{\ell m}$. Smooth angular data make these sums finite in the needed form domains.

For $\ell = 0$, the unitary sine transform diagonalizes h_0 . With hats denoting that transform,

$$\Gamma_{00} = \int_0^\infty \coth(\beta k/2) \left(k |\widehat{q}(k)|^2 + \frac{|\widehat{p}(k)|^2}{k} \right) dk, \quad (12)$$

$$E_{00} = \frac{1}{2} \int_0^\infty \left(k^2 |\widehat{q}(k)|^2 + |\widehat{p}(k)|^2 \right) dk. \quad (13)$$

2.2 Local interaction and exact channel

Let P be a degenerate pointer qubit with pointer observable Z_P . We couple it to the physical scalar by

$$S_{\text{int}} = - \int_{\mathcal{M}} \sqrt{-g} J(X) \phi(X) Z_P d^4 X, \quad J \in C_c^\infty(\mathcal{M}; \mathbb{R}). \quad (14)$$

The switching function and spatial smearing are both included in J . Since the commutator of two smeared free fields is a c-number, every Magnus term beyond second order vanishes. Up to a source-dependent scalar phase,

$$U_J = e^{i\Xi[J]} e^{-i\phi(J)Z_P}. \quad (15)$$

This standard exact evolution is reviewed in [2, 3, 4]. The interaction is compactly supported, and its densities commute at spacelike separation because Z_P is fixed and the scalar field is microcausal. Nevertheless, the same finite-dimensional operator Z_P appears across the smearing. Generic spatially extended nonrelativistic detectors can violate causal factorization [11]; fully relativistic localized probe fields provide a systematic replacement framework [12]. We therefore treat equation (14) as the standard prescribed gapless-detector idealization, not as an autonomous pointer field or material actuator.

Let E denote the causal propagator and K the positive-frequency map for the static one-particle structure. Tracing the Bunch–Davies field state gives a pointer dephasing channel

$$\mathcal{D}_\kappa \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & \kappa b \\ \bar{\kappa} c & d \end{pmatrix}, \quad |\kappa| = e^{-\Gamma}, \quad \Gamma = 2 \langle KEJ, \coth(\beta h/2) KEJ \rangle. \quad (16)$$

The phase of κ is a pointer rotation and is irrelevant to the distance from complete dephasing. If $\mathcal{D}_0$ denotes complete dephasing, then

$$\epsilon_{\text{obs}} := \frac{1}{2} \|\mathcal{D}_\kappa - \mathcal{D}_0\|_\diamond = \frac{1}{2} |\kappa| = \frac{1}{2} e^{-\Gamma}. \quad (17)$$

The equality follows from $\mathcal{D}_\kappa - \mathcal{D}_0 = (\kappa/2)(\text{id} - \text{Ad}_{Z_P})$ after the correctable phase is removed, and it is saturated by a pointer superposition. The target $\mathcal{D}_0$ is the binary instance of the quantum-to-classical pointer channel used in the observer rule of Harlow, Usatyuk, and Zhao [1]. The error ϵ_{obs} measures this channel approximation only; it is not their closed-universe encoding error and is not identified with $e^{-S_{\text{ob}}}$.

After the source switches off, the two conditional field states are opposite coherent displacements. Their common excess Killing energy and excess renormalized stress are

$$E_K = \langle KEJ, hKEJ \rangle, \quad \Delta \langle T_{ab}^{\text{ren}} \rangle = T_{ab}[\phi_J]. \quad (18)$$

The state-independent Hadamard subtraction cancels in the difference, and the background one-point function vanishes. Equation (18) prices the post-switch scalar field generated by *the same* J as the channel. It does not include the stress of a clock, trap, or material device that realizes the prescribed coupling.

2.3 Causal support

Suppose J is supported inside the centered areal ball $r \leq a < R$ during a static-time interval of length T . Its optical radius is

$$\ell = R \operatorname{arctanh}(a/R). \quad (19)$$

Finite propagation in the optical metric implies that the final Cauchy data of every partial wave are supported in

$$0 \leq x \leq L, \quad L = \ell + T. \quad (20)$$

No spectral cutoff or hard spatial wall is introduced.

3 The sharp localization–energy theorem

Let P_L be multiplication by $(0, L)$ on $L^2(\mathbb{R}_+)$. We first record the zero-temperature inverse kernels and then sum the complete KMS kernel.

Lemma 3.1 (Compressed half-line inverses). *For $h_0 = \sqrt{-\partial_x^2}$ with the half-line Dirichlet condition,*

$$(P_L h_0^{-1} P_L)(x, y) = \frac{1}{\pi} \log \frac{x+y}{|x-y|}, \quad (21)$$

$$(P_L h_0^{-2} P_L)(x, y) = \min(x, y). \quad (22)$$

Moreover,

$$\|P_L h_0^{-1} P_L\| \leq \frac{2 \operatorname{arsinh}(1)}{\pi} L, \quad (23)$$

$$\|P_L h_0^{-2} P_L\| = \frac{4}{\pi^2} L^2. \quad (24)$$

Proof. The unitary sine transform gives

$$\frac{2}{\pi} \int_0^\infty \frac{\sin(kx) \sin(ky)}{k} dk = \frac{1}{\pi} \log \frac{x+y}{|x-y|}$$

and

$$\frac{2}{\pi} \int_0^\infty \frac{\sin(kx) \sin(ky)}{k^2} dk = \min(x, y).$$

The logarithmic kernel is positive. With $u = x/L$, its row integral divided by L is

$$\frac{1}{\pi} [(1+u) \log(1+u) - 2u \log u - (1-u) \log(1-u)]. \quad (25)$$

Its maximum is $2 \operatorname{arsinh}(1)/\pi$ at $u = 1/\sqrt{2}$, so Schur's test proves equation (23). The kernel $\min(x, y)$ is the inverse of $-\partial_x^2$ on $(0, L)$ with Dirichlet condition at zero and Neumann condition at L . Its first eigenvalue is $(\pi/2L)^2$, proving equation (24). $\square$

Lemma 3.2 (Exact localized KMS kernel). *Set*

$$B_{\beta,L} := P_L h_0^{-1} \coth(\beta h_0/2) P_L. \quad (26)$$

Its integral kernel is

$$B_{\beta,L}(x, y) = \frac{1}{\pi} \log \frac{\sinh[\pi(x+y)/\beta]}{\sinh[\pi|x-y|/\beta]}. \quad (27)$$

Let $\tau = \pi L/\beta$ and define on $L^2(0, 1)$

$$k_\tau(u, v) := \frac{1}{\pi} \log \frac{\sinh[\tau(u+v)]}{\sinh[\tau|u-v|]}, \quad \Lambda(\tau) := \|\mathcal{K}_\tau\|. \quad (28)$$

Then $B_{\beta,L}$ is unitarily equivalent to $L\mathcal{K}_\tau$. $\mathcal{K}_\tau$ is positive, compact, and positivity improving. Its top eigenvalue $\Lambda(\tau)$ is simple and has a strictly positive eigenfunction.

Proof. Using $\coth z = 1 + 2 \sum_{n \geq 1} e^{-2nz}$ and integrating each sine-transform term gives

$$\begin{aligned} \frac{2}{\pi} \int_0^\infty \frac{\sin(kx) \sin(ky)}{k} \coth(\beta k/2) dk &= \frac{1}{\pi} \log \frac{x+y}{|x-y|} \\ &+ \frac{1}{\pi} \sum_{n \geq 1} \log \frac{(n\beta)^2 + (x+y)^2}{(n\beta)^2 + (x-y)^2}. \end{aligned} \quad (29)$$

Euler's product for $\sinh$ reduces this expression to equation (27). Scaling $x = Lu$, $y = Lv$ gives the unitary equivalence. The logarithmic diagonal singularity is square integrable on the unit square, so the operator is Hilbert–Schmidt. The spectral definition makes it positive, while the displayed kernel is strictly positive away from the measure-zero boundary and diagonal. The compact positivity-improving operator therefore has a simple positive top eigenfunction by the Perron–Frobenius–Jentzsch theorem. $\square$

The angular potential improves the complete thermal momentum form, not only the two estimates in lemma 3.1.

Lemma 3.3 (All-angular and coordinate domination). *For every $\ell \geq 0$, $A_\ell \geq A_0$. Moreover,*

$$\begin{aligned} h_\ell^{-1} \coth(\beta h_\ell/2) &= \frac{2}{\beta} A_\ell^{-1} + \frac{4}{\beta} \sum_{n \geq 1} [A_\ell + (2\pi n/\beta)^2]^{-1} \\ &\leq h_0^{-1} \coth(\beta h_0/2) \end{aligned} \quad (30)$$

as generalized quadratic forms. If q is supported in $[0, L]$, then

$$\|q\| \leq \frac{L}{\pi} \|h_\ell q\|, \quad (31)$$

$$\langle q, h_\ell q \rangle \leq \frac{L}{\pi} \|h_\ell q\|^2, \quad (32)$$

$$\frac{\Gamma_{q,\ell m}}{E_{q,\ell m} R} \leq \frac{2y}{\pi} + \frac{2y^2}{\pi^3}, \quad y = L/R, \quad \beta = 2\pi R. \quad (33)$$

Proof. The nonnegative angular potential gives $A_\ell \geq A_0$. The Mittag–Leffler expansion $\coth z/z = z^{-2} + 2 \sum_{n \geq 1} (z^2 + \pi^2 n^2)^{-1}$ gives equation (30); every resolvent is order decreasing. Extending q by zero places it in $H_0^1(0, L)$. Dirichlet Poincaré and $\|h_\ell q\|^2 = \langle q, A_\ell q \rangle$ give equation (31); Cauchy–Schwarz gives equation (32). Finally, $\coth z \leq 1 + z^{-1}$ yields

$$\Gamma_{q,\ell m} \leq \left[\frac{L}{\pi} + \frac{2L^2}{\beta\pi^2} \right] \|h_\ell q\|^2. \quad (34)$$

Divide by $E_{q,\ell m} = \|h_\ell q\|^2/2$ and set $\beta = 2\pi R$. No locality of the nonlocal vector $h_\ell q$ is assumed. $\square$

We can now state the sharp cost, rather than only an upper bound. The supremum below is unchanged if smooth compact data are replaced by their finite-energy closure.

Theorem 3.4 (Sharp compact-support dephasing cost). *Let*

$$C_{\text{opt}}(y) := \sup \left\{ \frac{\Gamma}{E_K R} : 0 < E_K < \infty, \text{supp}(q, p) \subset [0, yR] \right\}. \quad (35)$$

For the full conformally coupled massless scalar in the Bunch–Davies state, including arbitrary angular data,

$$\boxed{C_{\text{opt}}(y) = 2y\Lambda(y/2)}. \quad (36)$$

The supremum lies in the s-wave momentum sector. Its normalized optimizer is the unique positive top eigenfunction of equation (28), up to an overall sign. The following fully explicit bounds hold:

$$\max \left\{ \frac{3y}{\pi}, \frac{8y^2}{\pi^3} \right\} \leq C_{\text{opt}}(y) \leq F(y), \quad (37)$$

$$F(y) = \frac{4 \operatorname{arsinh}(1)}{\pi} y + \frac{8}{\pi^3} y^2. \quad (38)$$

If

$$M(\tau) := \max_{0 < u < 1} \int_0^1 k_\tau(u, v) dv, \quad (39)$$

then the sharper Schur bound $C_{\text{opt}}(y) \leq 2yM(y/2)$ holds. The maximum is attained at

$$u_\tau = \frac{1}{\tau} \operatorname{arsinh} \left(\frac{\sinh \tau}{\sqrt{2}} \right). \quad (40)$$

Finally, the exact support asymptotics obey the global remainder bounds

$$0 \leq C_{\text{opt}}(y) - 2y\Lambda(0) \leq \frac{y^3}{6\pi}, \quad (41)$$

$$0 \leq C_{\text{opt}}(y) - \frac{8y^2}{\pi^3} \leq \pi. \quad (42)$$

Thus temperature first corrects the localized vacuum cost at cubic order, while $8y^2/\pi^3$ is the sharp leading large-support coefficient.

Proof. For momentum data in the s-wave, lemma 3.2 gives

$$\sup_{p \neq 0} \frac{\Gamma_p}{E_p R} = \frac{2}{R} \|B_{\beta, L}\| = 2y\Lambda(y/2). \quad (43)$$

Equation (30) shows that no higher angular momentum sector is worse. The linear trial profile in equation (52) gives the lower bound $3y/\pi$. Also $\coth z \geq z^{-1}$ and equation (24) give

$$\sup_{p \neq 0} \frac{\Gamma_p}{E_p R} \geq \frac{8y^2}{\pi^3}. \quad (44)$$

The right side of equation (33) is bounded by $3y/\pi$ when $y \leq \pi^2/2$, and by $8y^2/\pi^3$ when $y \geq \pi^2/3$. These ranges overlap, so the entire coordinate sector is dominated by the s-wave momentum supremum for every $y > 0$. The domination is strict for nonzero coordinate data. Indeed, at finite β , the thermal momentum form is strictly larger than its vacuum part and strictly larger than $(2/\beta)A_0^{-1}$ on every nonzero vector. Applying those two strict inequalities to the trial profiles used above covers the same two overlapping ranges.

The angular domination is also strict for $\ell > 0$. For any $c > 0$, the resolvent difference $(A_0 + c)^{-1} - (A_\ell + c)^{-1}$ has a strictly positive quadratic form on nonzero data: equality would force $V_\ell^{1/2}(A_0 + c)^{-1}p = 0$, where $V_\ell = \ell(\ell + 1)/[R^2 \sinh^2(x/R)] > 0$ almost everywhere, and hence $p = 0$. The $n = 1$ term of equation (30) therefore excludes a higher-angular-momentum tie. Together with the simple top eigenvalue of $\mathcal{K}_{y/2}$, this proves the asserted uniqueness as well as equation (36) and its lower bounds.

For the first explicit upper bound, use $\coth z \leq 1 + z^{-1}$, lemmas 3.1 and 3.3, and the factor $1/2$ in the energy to obtain

$$\Gamma \leq E_K R F(y). \quad (45)$$

Positivity of k_τ and Schur's test give the $M(\tau)$ bound. If $f(s) = \log \sinh(\tau s)$, differentiation of the row integral gives $[f(1+u) + f(1-u) - 2f(u)]/\pi$. The identity

$$\sinh[\tau(1+u)] \sinh[\tau(1-u)] = \sinh^2 \tau - \sinh^2(\tau u) \quad (46)$$

then gives the unique stationary point equation (40).

For small support, write $k_\tau = k_0 + d_\tau$. Euler's product implies, for $A \geq B \geq 0$,

$$0 \leq \log \frac{\sinh A/A}{\sinh B/B} \leq \frac{A^2 - B^2}{6}. \quad (47)$$

Here $A = \tau(u+v)$ and $B = \tau|u-v|$, so Schur's test gives $0 \leq \Lambda(\tau) - \Lambda(0) \leq \tau^2/(3\pi)$. With $\tau = y/2$ this is equation (41).

For large support, write

$$k_\tau(u, v) = \frac{2\tau}{\pi} \min(u, v) + r_\tau(u, v). \quad (48)$$

The remainder is nonnegative in operator order by the resolvent expansion. Using $\int_0^\infty -\log(1 - e^{-s}) ds = \pi^2/6$ in its explicit row integral gives $\|r_\tau\| \leq \pi/(4\tau)$. Since the norm of the $\min(u, v)$ operator is $4/\pi^2$, multiplication by $2y = 4\tau$ proves equation (42). $\square$

3.1 Numerical realization of the sharp coefficient

The compact-operator formula can be evaluated directly without entering the proof. We use normalized piecewise constants on a uniform partition of $(0, 1)$. Cell integrals of the vacuum kernel are evaluated exactly from

$$\frac{d^2}{dz^2} \left(\frac{z^2}{2} \log |z| - \frac{3z^2}{4} \right) = \log |z|, \quad (49)$$

while Gauss–Legendre quadrature is applied only to the smooth thermal correction $\log[\sinh z/z]$. Figure 1 shows the resulting 128-cell Rayleigh–Ritz curve and positive top eigenfunctions.

At $y = 0.1, 1, 10$ the computed costs are respectively 0.0997264, 1.02960, and 27.7170. Across those three points, the largest relative change from 64 to 128 cells is 5.0×10^{-5} ; changing the thermal quadrature order from 8 to 12 at 64 cells changes the values by at most 1.1×10^{-15} . Every one of the nineteen plotted estimates lies inside equation (37) and below the exact-kernel Schur bound. Because the thermal quadrature is not interval enclosed, the plotted curve is numerical evidence rather than an additional rigorous lower bound; none of the theorem statements depends on it.

Combining equations (17), (20) and (36) gives the operational form.

Corollary 3.5 (Source-radius and duration bound). *If the source in equation (14) is supported in $r \leq a < R$ for a static-time interval of length T , set*

$$y = \operatorname{arctanh}(a/R) + T/R. \quad (50)$$

Then

$$\log \frac{1}{2\epsilon_{\text{obs}}} \leq E_K R C_{\text{opt}}(y) = 2E_K R y \Lambda(y/2) \leq E_K R F(y). \quad (51)$$

The first coefficient is sharp for the relaxed problem in which only the final Cauchy support L is fixed. For a cylinder with smaller spatial radius $\ell = L - T$, the bound remains valid, but reachability of its top eigenfunction is not asserted. Exact complete dephasing at finite a , T , and R requires $E_K = \infty$.

Remark 3.6. The final statement is a no-go only for exact complete dephasing in the frozen local scalar model. It is not a universal energy cost for quantum measurement, and it does not price the external device used to realize J .

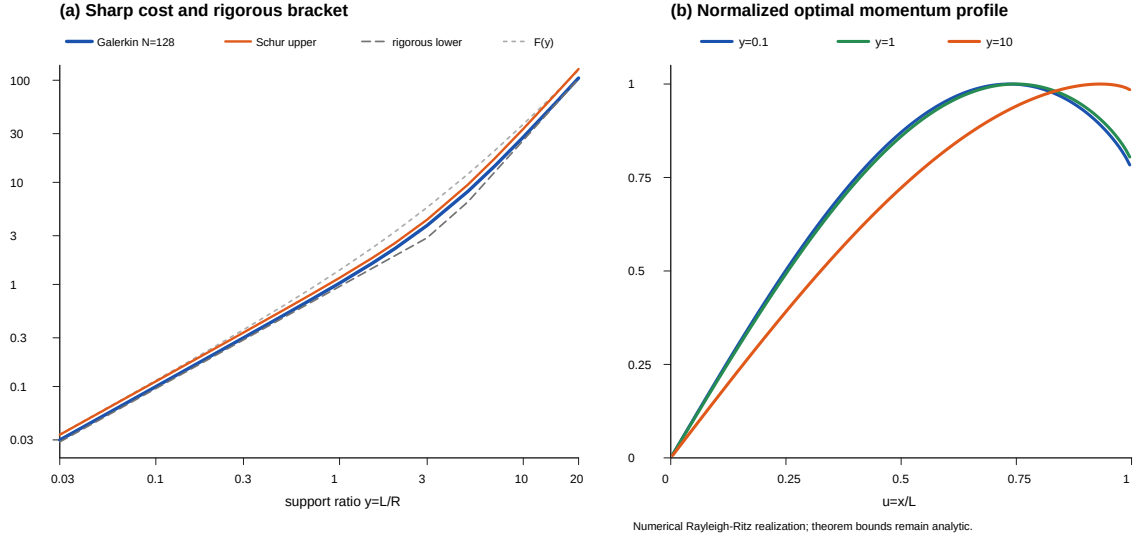


Figure 1: Numerical realization of the exact compact-kernel theorem. Panel (a) compares the 128-cell Galerkin estimate with the rigorous analytic lower bound, the exact-kernel row-Schur upper bound, and the simpler closed-form upper bound $F(y)$. Panel (b) shows the positive top eigenfunction, normalized by its maximum, at three support ratios.

4 Optimal profiles and smooth source realization

4.1 Explicit trial profiles

The exact optimizer in theorem 3.4 is the top eigenfunction of $\mathcal{K}_{y/2}$. Two elementary profiles give the explicit lower bounds in equation (37) and make both asymptotic regimes concrete. For the vacuum-controlled regime, set $L = yR$, choose $q = 0$, and use

$$p_L(x) = \sqrt{\frac{3}{L^3}} x \mathbf{1}_{(0,L)}(x). \quad (52)$$

The vacuum part of the thermal form is

$$\begin{aligned} \langle p_L, h_0^{-1} p_L \rangle &= \frac{3}{\pi L^3} \int_0^L \int_0^L xy \log \frac{x+y}{|x-y|} dx dy \\ &= \frac{3L}{2\pi}. \end{aligned} \quad (53)$$

For the second equality, scale $x = Lu$, $y = Lv$, split the unit square across the diagonal, and use

$$\int_0^1 t \log \frac{1+t}{1-t} dt = 1. \quad (54)$$

Since $\coth(\beta h_0/2) \geq 1$, the full thermal dephasing exponent is at least the value in equation (53). On the other hand, $E_K = 1/2$ for the normalized datum. Thus $\Gamma/(E_K R) \geq 3L/(\pi R) = 3y/\pi$.

For the thermal regime, use instead the normalized first eigenfunction of the Dirichlet–Neumann Green operator,

$$p_L^{\text{th}}(x) = \sqrt{\frac{2}{L}} \sin \frac{\pi x}{2L} \mathbf{1}_{(0,L)}(x). \quad (55)$$

Since $\coth z \geq z^{-1}$ and $\langle p_L^{\text{th}}, h_0^{-2} p_L^{\text{th}} \rangle = 4L^2/\pi^2$, this datum gives

$$\frac{\Gamma}{E_K R} \geq \frac{8y^2}{\pi^3}. \quad (56)$$

The first profile and the vacuum Schur bound also show

$$\frac{3}{\pi} \leq \liminf_{y \downarrow 0} \frac{C_{\text{opt}}(y)}{y} \leq \limsup_{y \downarrow 0} \frac{C_{\text{opt}}(y)}{y} \leq \frac{4 \operatorname{arsinh}(1)}{\pi}. \quad (57)$$

The ratio of the analytic upper and lower linear coefficients is $4 \operatorname{arsinh}(1)/3 \simeq 1.17516$.

For a target exponent $\Gamma_\star$, scaling equation (52) by an amplitude A yields the sufficient energy bound

$$E_K \leq \frac{\pi \Gamma_\star}{3L}. \quad (58)$$

This is an upper bound on the energy needed by this simple family. The sharp infimum over all supported data is instead $\Gamma_\star/[RC_{\text{opt}}(L/R)] = \Gamma_\star/[2L\Lambda(L/2R)]$.

4.2 Smooth compact sources

The hard endpoint in equation (52) is useful for exact constants but is not a smooth source. We now show that it is a legitimate limiting profile and that every strict window persists.

Lemma 4.1 (Smooth density closure). *Fix an optical source radius $\ell > 0$ and a normalized $p \in L^2(0, \ell)$. There are normalized $p_n \in C_c^\infty([0, \ell])$, regular at the origin, such that*

$$p_n \longrightarrow p \quad \text{in } L^2(\mathbb{R}_+). \quad (59)$$

Their thermal dephasing forms converge, and their cumulative energy measures converge uniformly.

Proof. Density of smooth functions supported strictly inside $(0, \ell)$ gives equation (59); such functions are automatically regular radial data at the origin. On data supported in $[0, \ell]$,

$$P_\ell h_0^{-1} \coth(\beta h_0/2) P_\ell \leq P_\ell h_0^{-1} P_\ell + \frac{2}{\beta} P_\ell h_0^{-2} P_\ell \quad (60)$$

as quadratic forms. The right side is bounded by lemma 3.1; hence the covariance is continuous under L^2 convergence. Finally,

$$\sup_{0 \leq x \leq \ell} \left| \int_0^x (p_n^2 - p^2) \, ds \right| \leq \|p_n - p\|_2 (\|p_n\|_2 + \|p\|_2), \quad (61)$$

which proves uniform convergence of cumulative energy. $\square$

It remains to produce a supported final datum by a smooth spacetime source. This also proves sharpness over sources when the source ball approaches the declared final-support ball; it does not prove exact controllability from a strictly smaller cylinder of optical radius $L - T$. Let $\phi_{\text{free},n}$ be the homogeneous solution with final data $(q, p) = (0, p_n)$. Choose a smooth time cutoff η_n that vanishes before the protocol, equals one near the final slice, and changes over a time shorter than the optical gap between $\text{supp } p_n$ and ℓ . If $\mathcal{P}$ is the conformal wave operator, set

$$J_n = \mathcal{P}(\eta_n \phi_{\text{free},n}). \quad (62)$$

The retarded solution sourced by J_n is $\eta_n \phi_{\text{free},n}$. It therefore has exactly the desired final data. Finite propagation keeps J_n inside the declared optical worldtube. This construction proves smooth source existence, but it does not bound the stress of an autonomous mechanism that generates the prescribed J_n .

5 Flux-free Einstein–scalar constraint data

The all-angular theorem does not require spherical symmetry. We now restrict to the s-wave and to a final slice on which the radial field coordinate vanishes. This subclass is useful because its momentum density vanishes and its energy measure has a direct constraint interpretation.

Let $u(t, x)$ be the normalized rescaled s-wave. The physical conformal scalar is

$$\phi(t, r) = \frac{u(t, x(r))}{\sqrt{4\pi} r}. \quad (63)$$

On a final slice with $u = 0$ and $p = \partial_t u$, the physical normal derivative in the de Sitter geometry is

$$v(r) := n^a \nabla_a \phi = \frac{p(x(r))}{\sqrt{4\pi} r \sqrt{N(r)}}. \quad (64)$$

For the conformally coupled stress tensor, every improvement term containing ϕ or a spatial derivative of ϕ vanishes on this slice. Directly,

$$\rho = T_{ab} n^a n^b = \frac{1}{2} v^2, \quad j_i = -T_{ab} n^a h^b_i = 0. \quad (65)$$

Since $dr = N dx$, equations (64) and (65) imply the exact measure identity

$$4\pi r^2 \rho(r) dr = \frac{1}{2} p(x)^2 dx. \quad (66)$$

Use the same numerical function v as the scalar normal-momentum datum on a new spherical initial slice, and consider gravitational data with vanishing extrinsic curvature,

$$dl^2 = \frac{dr^2}{f(r)} + r^2 d\Omega_2^2, \quad K_{ij} = 0, \quad f(r) = N(r) - \frac{2Gm(r)}{r}. \quad (67)$$

The momentum constraint is satisfied by equation (65). The Hamiltonian constraint with cosmological radius R is solved by

$$m'(r) = 4\pi r^2 \rho(r), \quad m(0) = 0. \quad (68)$$

Absorbing the Bunch–Davies vacuum expectation into the background cosmological constant, the coherent excess stress in equation (18) supplies precisely the classical source above. Although $K_{ij} = 0$, the nonzero scalar normal momentum means that the full Einstein–scalar data are not invariant under time reflection. Equation (66) then gives

$$m(b) = E_K \quad (69)$$

for any areal radius b outside the field support.

Define the dimensionless radial constraint ratio

$$Q(r) = \frac{2Gm(r)}{rN(r)}. \quad (70)$$

For a comparison ball $0 < r \leq b < R$, set

$$Q_b := \sup_{0 < r \leq b} Q(r). \quad (71)$$

If $Q_b \leq \delta < 1$, then throughout that ball

$$f(r) \geq N(r)(1 - \delta) > 0, \quad 0 \leq \frac{g_{rr}}{g_{rr}^{\text{dS}}} - 1 \leq \frac{\delta}{1 - \delta}. \quad (72)$$

Corollary 5.1 (Necessary final-slice constraint cost). *Let the hypotheses of corollary 3.5 hold and suppose the source is engineered to finish with spherical $q = 0$ data. Set*

$$b = R \tanh(L/R), \quad L = R \operatorname{arctanh}(a/R) + T. \quad (73)$$

Then

$$Q_b \geq Q(b) \geq \frac{2G}{bN(b)RC_{\text{opt}}(L/R)} \log \frac{1}{2\epsilon_{\text{obs}}}. \quad (74)$$

Equivalently, a local budget $Q_b \leq \delta < 1$ implies

$$\epsilon_{\text{obs}} \geq \frac{1}{2} \exp \left[-\frac{\delta bN(b)R}{2G} C_{\text{opt}}(L/R) \right]. \quad (75)$$

Proof. Outside the support, equation (69) gives $Q(b) = 2GE_K/[bN(b)]$. Apply the sharp energy lower bound in corollary 3.5; solving the same inequality in the other direction gives equation (75). $\square$

The compact mass shifts the cosmological horizon relative to the pure de Sitter value R . Accordingly, Q_b is a local comparison norm on the ball containing the final datum; no global smallness claim relative to the unperturbed lapse is intended.

5.1 An explicit local weak-constraint window

For the source optical radius $\ell = R \operatorname{arctanh}(a/R)$, take the ideal profile

$$q = 0, \quad p(x) = A \sqrt{\frac{3}{\ell^3}} x \mathbf{1}_{(0,\ell)}(x). \quad (76)$$

Its cumulative mass is

$$m(x) = E_K \left(\frac{x}{\ell} \right)^3. \quad (77)$$

Inside the support, the function

$$\frac{\operatorname{arctanh}(r/R)^3}{r[1 - r^2/R^2]} \quad (78)$$

is strictly increasing on $(0, a]$. One way to see this is to factor it as $(x/r)(x^2/N)$; both factors increase with r . Hence the wall value controls the complete profile. Combining equation (58) with the wall constraint gives the sufficient estimate

$$Q_a \leq \frac{2\pi G \Gamma_\star}{3\ell a[1 - a^2/R^2]}. \quad (79)$$

For illustration, set

$$\epsilon_{\text{obs}} = 10^{-6}, \quad a/R = 0.2, \quad T/R = 0.1, \quad G/R^2 = 10^{-6}, \quad \delta = 0.25. \quad (80)$$

The universal theorem requires

$$C_{\text{opt}}(L/R) \leq 0.340766570374, \quad E_K R \geq 38.5083647231, \quad Q(b) \geq 2.86896268303 \times 10^{-4}, \quad (81)$$

whereas the constructive family has

$$E_K R \leq 67.7824380936, \quad Q_a \leq 7.06067063475 \times 10^{-4}. \quad (82)$$

Thus the fixed numerical point has a nonempty local weak-constraint window. By lemma 4.1, smooth compact profiles preserve this strict margin.

The construction closes only the final Hamiltonian and momentum constraints. It does not specify the lapse, evolve the Einstein–scalar system, include the source actuator during switching, or recompute the quantum channel in the perturbed geometry.

6 Comparison with local entropy and horizon decoherence

6.1 Bekenstein entropy is a different quadratic form

For a Klein–Gordon wave packet localized in a spatial region of half-width a , Hollands, Longo, and Morsella prove the Bekenstein-type inequality

$$S(\Phi | B) \leq 2\pi a E, \quad (83)$$

where $S(\Phi | B)$ is the entropy quadratic form of the local standard subspace [10]. At second quantization this is the relative entropy of the corresponding coherent excitation. This powerful result is directly adjacent to theorem 3.4, but it controls a different functional.

We make the distinction explicit in the simplest common setting. Consider the massless conformal vacuum in four-dimensional Minkowski space, restrict to radial $q = 0$ data in a ball of radius a , and normalize $\|p\|_2 = 1$. The conformal-ball modular Hamiltonian and the half-line dephasing kernel give

$$S_B[p] = \frac{\pi}{2a} \int_0^a (a^2 - x^2) p(x)^2 dx, \quad (84)$$

$$\Gamma_0[p] = \langle p, h_0^{-1} p \rangle = \frac{1}{\pi} \int_0^a \int_0^a p(x) p(y) \log \frac{x+y}{|x-y|} dx dy. \quad (85)$$

Proposition 6.1 (Quadratic-form separation). *At fixed a , there is no finite constant c such that $\Gamma_0[p] \leq c S_B[p]$ for all smooth normalized radial momenta, and no finite constant c' such that $S_B[p] \leq c' \Gamma_0[p]$ for all such data.*

Proof. Let $f \in C_c^\infty(0, 1)$ be positive and normalized in $L^2(0, 1)$. For a packet in a boundary layer, set

$$p_\delta^{\text{wall}}(x) = \delta^{-1/2} f\left(\frac{a-x}{\delta}\right). \quad (86)$$

Substitution in equation (84) gives

$$S_B[p_\delta^{\text{wall}}] = \pi \delta \int_0^1 u f(u)^2 du + O(\delta^2). \quad (87)$$

For the logarithmic kernel, scaling both variables gives

$$\Gamma_0[p_\delta^{\text{wall}}] = \frac{\delta}{\pi} \left(\int_0^1 f(u) du \right)^2 \log \frac{1}{\delta} + O(\delta). \quad (88)$$

Thus $\Gamma_0/S_B \rightarrow \infty$.

For a packet near the center, set

$$p_\delta^{\text{ctr}}(x) = \delta^{-1/2} f(x/\delta). \quad (89)$$

Then

$$S_B[p_\delta^{\text{ctr}}] = \frac{\pi a}{2} + O(\delta^2), \quad \Gamma_0[p_\delta^{\text{ctr}}] = O(\delta), \quad (90)$$

because the logarithmic kernel becomes scale invariant after $x = \delta u$, $y = \delta v$, while the two integration measures leave one factor of δ . Hence $S_B/\Gamma_0 \rightarrow \infty$. $\square$

The separation has a simple physical meaning. The Bekenstein entropy is a local modular-information functional whose conformal weight vanishes at the ball boundary. The zero-temperature dephasing exponent is a global one-particle norm, equivalently an entangling-particle count for the exact detector model. At finite temperature, the factor $\coth(\beta h/2)$ enhances infrared fluctuations further. Thermal dephasing can increase even while the distinguishability of two displaced thermal states decreases. The two notions of information should not be identified.

6.2 Relation to DSW and exact detector work

DSW decoherence describes the loss of coherence of superposed charges or masses near Killing horizons through soft field modes [5, 7]. Gralla and Wei gave a general horizon formulation and precise rates in electromagnetic and Klein–Gordon analogues [6]; Li evaluated the local scalar, electromagnetic, and gravitational cases in de Sitter [8]. These works establish the physical importance of the low-frequency KMS sector that appears in equation (10).

Our setup differs in its question. We do not derive a new horizon decoherence rate. We start from the exact pointer channel and maximize its dephasing functional subject to finite final support and post-switch field energy. The result is consequently a constraint on a local implementation, not a claim that horizon modes require a large Killing energy. Indeed, sufficiently soft horizon radiation can carry negligible Killing energy when its support and duration are allowed to grow. In equation (51), that escape route is visible in the growth of C_{opt} with $L = \ell + T$.

Barcellos and Landulfo already separate field background energy, individual switching work, and communication work for the exact gapless-detector channel [3]. We use their same exact channel class. The new step is to eliminate the source amplitude, identify the exact compact KMS kernel, and solve the full final-support optimization. We therefore make no novelty claim for Magnus termination, the KMS characteristic function, the coherent stress identity, or switching-energy accounting.

Likewise, spectral extremization of Riesz-potential operators is not new [9]. In the zero-temperature limit our reflected logarithmic kernel belongs to that broad potential-theoretic family. The claim here is not a first eigenvalue theorem for a positive compact kernel; it is the identification of the exact thermal operator governing this local pointer channel, proof that it controls the complete canonical and angular phase space, and extraction of the sharp physical support laws.

7 Discussion and claim boundary

We have solved the sharp coupling-free localization–energy optimization for an exact local pointer channel in the de Sitter static patch. Three features are worth separating.

First, the channel is nonperturbative but not new. Its value here is that the same compact source determines the pointer error, the coherent field stress, the Killing-energy excess, and the causal support. This prevents a resource ledger assembled from unrelated effective models.

Second, compact support is the decisive hypothesis. The static-patch KMS state has arbitrarily soft modes. Without a support or duration restriction, no energy gap controls the dephasing exponent. Localization replaces that missing gap by a compact thermal inverse operator. The exact formula equation (36) and remainder bounds equations (41) and (42) show that both the linear small-support law and quadratic large-support law are intrinsic to the model.

Third, the gravity result is a final-slice statement. For the flux-free spherical subclass, the conformal field map identifies the optical energy measure with the exact Einstein–scalar constraint mass measure. This is stronger than assigning an unrelated spherical mass profile, but weaker than deriving the complete detector protocol in semiclassical gravity.

7.1 What is and is not established

The results establish:

1. the exact reduced pointer channel and same-source post-switch field energy and renormalized-stress excess;

2. the exact all-angular optimum equation (36), its unique s-wave momentum sector, and a valid source-radius and duration envelope;
3. sharp small- and large-support asymptotics and smooth compact source sequences approaching the optimal Rayleigh quotient;
4. exact flux-free spherical Einstein–scalar Hamiltonian and momentum constraint data and a nonempty local weak-constraint example; and
5. an explicit proof that the dephasing exponent and localized Klein–Gordon Bekenstein entropy are incomparable quadratic forms.

They do not establish:

1. the stress, clock energy, or size of an autonomous actuator that realizes the prescribed source;
2. an autonomous relativistic pointer field whose local degrees of freedom reproduce the single smeared qubit operator Z_P ;
3. the pointer channel on the perturbed geometry, a lapse solution, or a coupled Einstein–scalar evolution through the switching interval;
4. a universal energy cost of measurement, observer complementarity, a de Sitter holographic dictionary, or ER=EPR;
5. a new exact detector channel, a new DSW decoherence mechanism, or the first localization–information–energy bound in QFT;
6. a closed elementary formula for $\Lambda(\tau)$ at finite support; or
7. exact controllability of the sharp final-support optimizer from every strictly smaller source cylinder with the same causal output radius.

7.2 Next mathematical and physical questions

The sharp constant is the simple top eigenvalue in equation (28). The coordinate comparison proves that no second optimization is missing. Figure 1 supplies a converged numerical realization and optimal-profile plot. A future interval-certified discretization could turn those estimates into tighter rigorous finite- y brackets, but it would not change the theorem.

The most important physical extension is perturbative stability of the channel under the final metric distortion. A controlled result would compare the Bunch–Davies two-point function and propagator on the pure de Sitter background with those on the small- Q Einstein–scalar development, while including a conserved model for the switching apparatus. Until then, the constraint result should remain a corollary rather than the title claim.

A complementary extension is to replace the shared smeared qubit by a localized relativistic probe field, following the detector-from-field framework of [12], and then ask whether the exact localization–energy optimization survives the additional probe modes and their stress tensor.

A second extension is geometric. The proof uses a centered ball and the positivity of the de Sitter angular potential. The operator formulation suggests an analogue on other static spacetimes in terms of localized norms of h^{-1} and h^{-2} . Such a generalization would need to preserve the exact channel/stress relation and to state carefully which energy is generated by the local coupling.

Finally, the comparison in proposition 6.1 suggests that local modular entropy and environment-induced dephasing probe complementary aspects of localization. Establishing useful inequalities after adding a

boundary-distance constraint, a restricted source family, or a finite resolution could connect the operational detector problem more directly to modular-energy methods.

7.3 Reproducibility

The constants and illustrative window are reproduced from the repository root by

```
PYTHONPATH=. python experiments/local_scalar_observer_cost_audit.py
PYTHONPATH=. python experiments/local_scalar_observer_spectrum.py
PYTHONPATH=. python -m pytest -q tests/test_local_scalar_observer_cost.py
```

The frozen JSON certificate records the source hashes, theorem status, exact coefficient decomposition, energy and constraint bounds, all-angular lift, smooth-source density argument, and Bekenstein comparison. The executable record checks algebra and provenance; it does not replace the proofs above or certify literature novelty. The separate spectrum JSON records the Galerkin method, convergence and quadrature comparisons, plotted values, optimizer profiles, and figure hash.

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